

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No-	Team No-	Project Title-
Team Lead Roll Number		2520030103
Team Lead Name		K. Poojitha Sai
Team Member Roll Number		2520030501
Team Member Name		Anushka Kulkarni

1. Abstract

The proposed system aims to develop a Blood Bank Management System that helps manage blood inventory, donor records, issue requests, and blood availability efficiently. Traditional blood bank systems often rely on manual records, which can lead to delays, incorrect inventory updates, and difficulty in tracking blood availability and donor information.

This project provides a centralized digital platform where users and administrators can manage donor profiles, monitor blood stock by group and component, and process blood requests quickly. The system includes compatibility checking, donation history tracking, expiry alerts for blood units, and notifications for important updates.

The application is developed using modern front-end technologies such as React.js, HTML5, CSS3, and JavaScript (ES6+). It follows component-based architecture with reusable UI components and modular frontend design. API handling and asynchronous updates are implemented using Axios and REST APIs.

An admin panel is included for managing donor data, inventory records, and request approvals. The system improves efficiency, reduces manual work, enhances data organization, and ensures better accessibility and user experience in healthcare blood management services.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Donor Profile, Donation History, Notifications	2520030103	
UI/UX & Frontend Integration	2520030103	
Inventory Table, Compatibility Check, Issue Request	2520030501	
Expiry Alert, Admin Panel, Postman API	2520030501	

3. Module Details

1. Donor Profile

Stores donor personal details, blood group information, and contact details.

2. Donation History

Maintains records of previous donations and donor activity.

3. Notifications

Displays alerts and notifications related to blood requests and updates.

4. UI/UX & Frontend Integration

Ensures responsive design, reusable components, and smooth navigation.

5. Inventory Table

Displays available blood units categorized by blood group and component.

6. Compatibility Check

Checks compatibility between donor and recipient blood groups.

7. Issue Request

Handles blood request submissions and approval process.

8. Expiry Alert

Tracks blood expiry dates and displays alerts for expiring units.

9. Admin Panel

Allows administrators to manage donors, requests, and inventory records.

10. Postman API

Used for API testing and validating request/response handling.

4. Frameworks / Platforms / Software Used

1. Frontend: HTML5, CSS3, JavaScript (ES6+), React.js

2. Styling: Bootstrap / Tailwind CSS

3. Backend: Node.js, Express.js

4. Database: MySQL / LocalStorage

5. API Communication: REST APIs, Axios

6. Routing: React Router

7. State Management: Context API / Redux

8. Testing Tools: Postman



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9. Development Tools: Visual Studio Code, Git, Chrome DevTools

Guide Name and Signature (remarks if any) _____